DATA SCIENCE ASSIGNMENT | IMF COU NTRY REPORT ANALSIS

Financial Intelligence Pipeline: St. Kitts and Nevis

Source: IMF Country Report No. 24/126, 2024 Article IV Consultation-Press Release and Staff Report (May 2024). **Dataset: supplied** 61-page PDF. **Purpose:** extract marco-financial indicators, identify financial-stability themes and risks, and demonstrate grounded retrieval and visual analytics.

1. Introduction

This project analyses the IMF's 2024 Article IV assessment of St. Kitts and Nevis. The report describes a tourism-supported recovery alongside vulnerabilities associated with volatile Citizenship-by-Investment (CBI) receipts, a systemically important bank, pension liabilities and exposure to natural disasters. The pipeline combines PDF parsing, text cleaning, overlapping chunking, structured indicator extraction, lexical thematic scanning, dense-vector retrieval and dashboard visualisation. All numerical values below are tied to the report's tables or narrative; missing values are retained as null rather than imputed.

2. Methodology and architecture

The 61-page PDF is ingested with pypdf. Page-level extraction is retained to preserve source locations. Cleaning removes repeated copyright artefacts, normalises whitespace and joins hyphenated line breaks. The notebook then creates overlapping chunks of approximately 400 tokens with a 50-token overlap. A tiktoken tokenizer is used in the reproducible notebook; a whitespace fallback is included if tokenizer installation fails. Chunk metadata preserves the PDF for retrieval provenance.

The notebook contains generation prompts for executive and section summaries and a separate judge prompt for factual alignment, completeness, consistency and schema compliance. The generation model and judge model are configured independently. Live model calls require an API key; no LLM score is claimed in this draft because no API key was supplied for this execution. The notebook that state explicitly.

3. Source-grounded summary

• Real GDP growth slowed from 8.8% in 2022 to 3.4% in 2023 as public and private investment projects were delayed, despite strong tourism.

• Average consumer-price inflation reached 3.6% in 2023, driven by food, oil and shipping costs; IMF staff projected moderation to 2.5% in 2024 and 2.2% in 2025.

• The fiscal balance moved to a surplus of 1.0% of GDP in 2023 and public debt declined to 54.4% of GDP, helped by CBI revenues of about 21.5-22% of GDP.

• Fiscal sustainability remains exposed to volatile CBI receipts, which the IMF baseline projects to decline to 10% of GDP by 2029, contributing to a projected overall deficit of 3.9% of GDP.

• The banking system shows mixed signals: aggregate NPLs fell to 19.4% in 2023, but the systemic bank's NPL ratio was 39% in 2023 and sector Tier 1 capital fell to 6.8%.

4. Structured extraction and qualitative insights

The target schema contains six numerical fields plus country name. Table 1 (printed p.4) supplies the 2024 GDP growth forecast and average inflaction forecast. Table 5 (printed p.28) supplies banking-sector aggregate NPLs and Tier 1 capital. The report does not provided a signal country-specific central-bank policy rate because monetary plicy is managed by the ECCB within the ECCU framework. It does not provide a national umemployment point estimate; the report notes limitations in labour-market data. These two values are therefore null.

Requested field	Extracted value	Reference / interpretation
GDP growth forecast	4.7%	2024 real GDP growth, Table 1, p.4
Headline inflation	2.5%	2024 consumer prices, period average, Table 1, p.4
Central-bank policy rate	null	No country-specific policy-rate figure reported
NPL ratio	19.4%	2023 banking-sector NPLs / gross loans, Table 5, p.28
Unemployment rate	null	No national point estimate supplied; data limitations noted
Tier 1 capital ratio	6.8%	2023 regulatory Tier 1 / risk-weighted assets, Table 5, p.28

Important distinction: the 19.4% figure is the banking-sector aggregate. The narrative separately reports that the systemic bank's NPL ratio was 39% in 2023Q2 (printed p.5). These are different scopes and should not be conflated.

Growth drivers / strengths

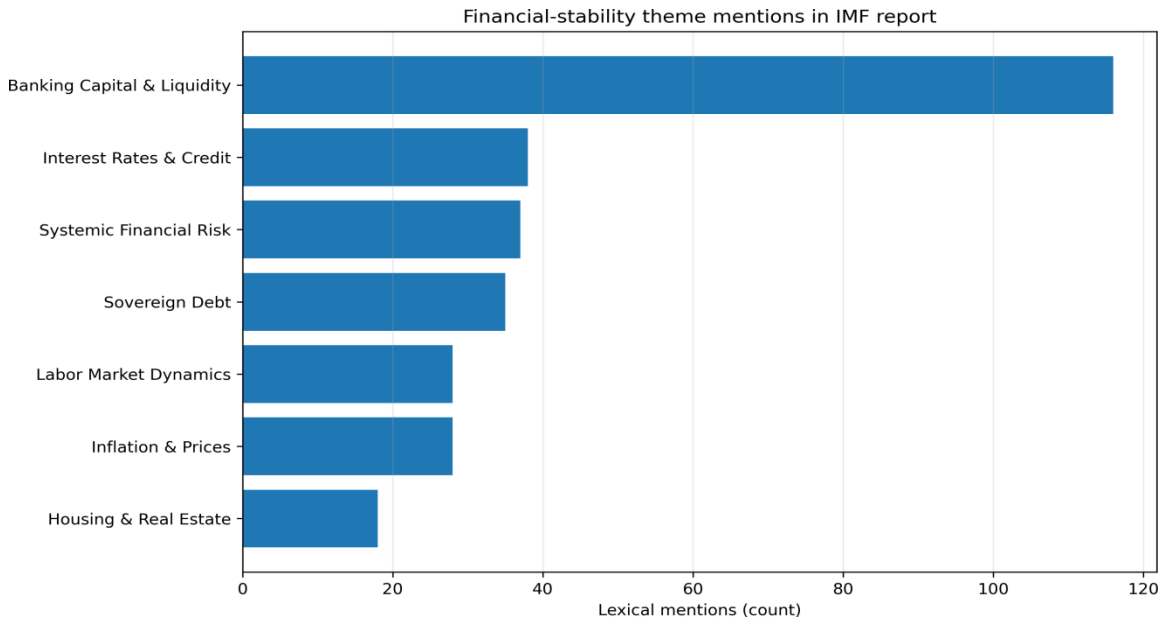
- Tourism recovery supported growth and narrowed the current-account deficit; toursim receipts continued to rise.
- Public debt fell to 54.4% of GDP in 2023 and remained below the ECCU 60% regional ceiling.
- International reserves were assessed as adequate.
- Planned solar and geothermal investments could reduse fossil-fuel imports, lower energy costs and improve competitiveness.
- CBI inflows supported the 2023 fiscal surplus and provided fiscal space, although their volatility creates a concentration risk.
- Banking-sector aggregate NPL ratio declined from 24.0% in 2019 to 19.4% in 2023.

Systemic vulnerabilities / challenges

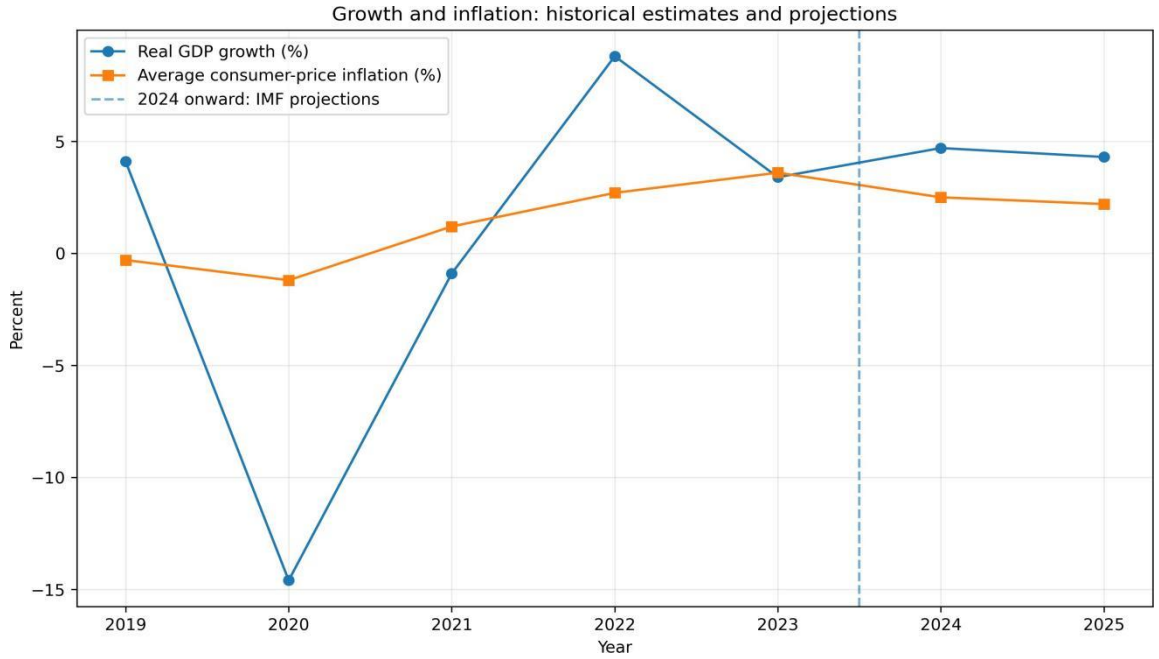
- High dependency on volatile Citizenship-by-Investment revenues; CBI receipts are projected to decline over the medium term.
- The systemic bank has weak asset quality: its NPL ratio was 39% in 2023Q2, alongside low capital and a large foreign investment portfolio funded by public deposits.
- Banking-sector Tier 1 capital ratio declined to 6.8% in 2023 from 19.4% in 2019.
- Social Security Fund faces an actuarial imbalance estimated around 300% of GDP over 60 years and reserves are projected to be depleted by 2040.
- High exposure to hurricanes and other natural disasters, alongside water-supply stress and infrastructure resilience needs.
- External risks include geopolitical distruption, commodity-price volatility and a slowdown in key tourism source markets.

5. Thematic distribution and trend analysis

The seven-theme scan uses case-insensitive keyword and phrase occurrence counts across the main report pages. Counts are reproducible but should be interpreted as lexical mentions, not semantic topic classification. Categories may overlap; for example, a paragraph about systemic-bank capital may contribute to both banking-capital and systemic-risk categories. A future enhancement could replace lexical counting with embedding-based or LLM-assisted classsification followed by human review.

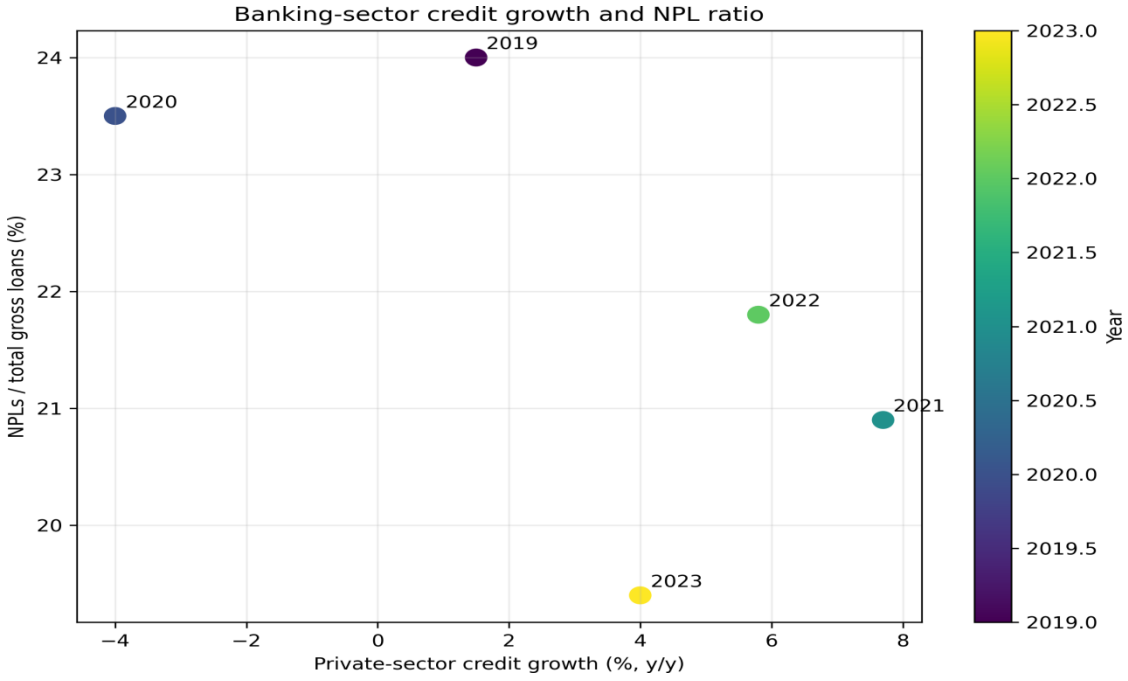


Banking capital and liquidity has the highest mention count in this lexical scan, reflecting the report's sustained discussion of the systemic bank, NPLs, capital restoration, public deposits and credit-union supervision. The count is not a measure of actual risk magnitude.

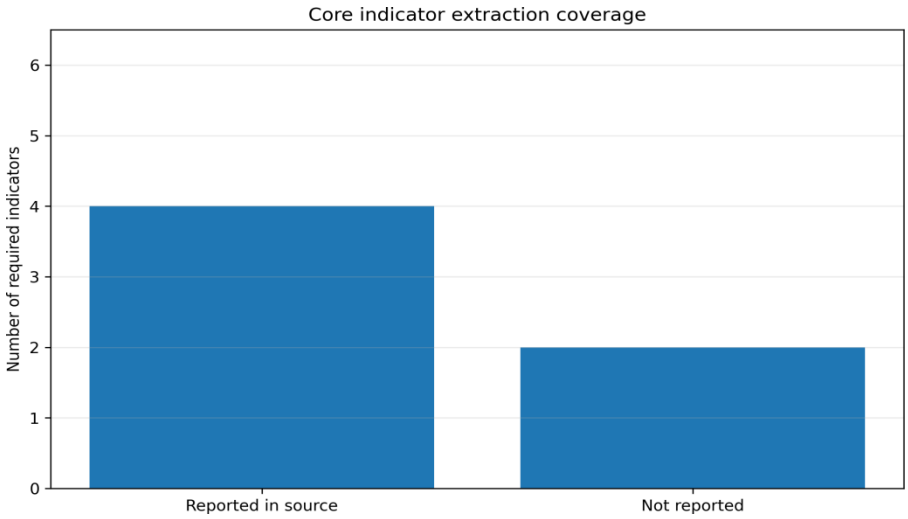


GDP growth recovered sharply after the 2020 contraction, then moderated in 2023. The IMF projected a rebound to 4.7% in 2024. Average inflation rose to 3.6% in 2023 and was projected to moderate to 2.5% in 2024 and 2.2.% in 2025 (Table 1, printed p.4)

6. Financial-sector diagnostics and retrieval



The scatter plot pairs annual private-sector credit growth with banking-sector NPLs for 2019-2023 using Table 5 (printed p.28). It is descriptive, not casual. Credit growth was negative in 2020 and recovered thereafter; the aggregate NPL ratio declined from 24.0% in 2019 to 19.4% in 2023. The separate-bank NPL ratio remains materially higher at 39% in 2023Q2.



Four of the six requested numerical indicators have explicit source values; two are unavailable in the requested form. The notebook includes an independent judge prompt loop and a FAISS retrieval pipeline. Dense embeddings are created locally with Sentence Transformers and indexed with FAISS IndexFlatL2. Multi-query retrieval expands a question into several related queries and deduplicates retrieved chunks.

RAG test questions

• What are the systemic bank's asset-quality and capital weaknesses?

• What fiscal reforms does the IMF recommend to reduce reliance on CBI receipts?

• How could solar and geothermal projects affect energy imports and growth?

The notebook returns retrieved chunk text with PDF page metadata. Answers should cite the retrieved evidence and be manually checked against the report. The notebook includes a baseline single-query versus multi-query retrieval comparison; retrieval quality scores should only be added after a human-checked evaluation set is run.

7. Evaluation, limitations and conclusion

Evaluation framework

Numerical extraction is checked against a manually verified reference map derived from Table 1 and Table 5. This deterministic validation is separate from an LLM-as-a-Judge evaluation. The second-model judge prompt is included in the notebook and is designed to score factual alignment, completeness, consistency and schema compliance from 0 to 5. The live judge has not yet been executed in this environment; the final submitted report should insert the actual raw judge output and discuss errors after running the notebook with an API key.

Limitations

First, keyword counts approximate thematic density and may count repeated mentions or overlapping concepts. Second, several indicators are not available in the report in the exact requested form. Third, the report's banking-sector aggregate NPL ratio should not be substituted for the systemic bank's distinct NPL ratio. Fourth, 2024 and 2025 values are projections, not realised outcomes. Fifth, dense retrieval can return relevant-looking but incomplete context; source-page checking remains necessary. The report itself notes weaknesses in labour-market and other statistical coverage.

Conclusion

The IMF report presents a positive medium-term opportunity from tourism and renewable-energy investment, but also identifies structural fiscal dependence on CBI receipts and financial-stability concerns around the systemic bank and credit unions. The extracted figures show that the sector-wide NPL ratio improved over 2019–2023 while Tier 1 capital weakened materially. The recommended policy direction combines fiscal-rule and tax reform, improved CBI governance, pension reform, renewable-energy investment and restructuring of the systemic bank. The pipeline makes these findings searchable, structured and visually interpretable while preserving missing-data transparency.